

# Free Space Optical link performance prediction based on NRBO-BP

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**Abstract**—Free space optical (FSO) communication offers high transmission rates and improved information security, but its performance can be significantly impacted by adverse weather. Recently, machine learning algorithms have been employed to enhance prediction accuracy and optimize network performance. This paper presents a performance analysis of the FSO system and proposes a novel Q-factor prediction method combining the Newton-Raphson-Based Optimizer (NRBO) with a Backpropagation (BP) neural network. NRBO optimizes the BP network's initial parameters, addressing challenges like local optima and slow convergence. The experimental results show that the NRBO-BP model outperforms other models, demonstrating a higher coefficient of determination ( $R^2$ ) and lower mean absolute error (MAE).

**Keywords**- Free Space Optics; link performance; prediction; Q-factor; BP neural network

## I. INTRODUCTION

As wireless communication technology advances rapidly, Free Space Optical (FSO) Communication, as an efficient optical wireless communication technology with unlicensed spectrum, has gradually become a strong complement to fiber optic communication[1]. FSO technology enables high-speed data communication through light wave transmission in free space, offering great potential, especially for medium-to-short range communication. Compared to optical fiber systems, FSO is more cost-effective, flexible, and provides similar bandwidth. However, its performance is affected by environmental factors such as weather changes, atmospheric attenuation, and turbulence, which can degrade signal quality and link performance.[3].

The Quality Factor (Q-factor) is an essential metric for evaluating the link performance of an FSO system, indicating both the signal quality and the system's transmission performance. A higher Q value indicates better signal quality[5]. Under adverse environmental conditions, the Q-factor of FSO systems undergoes significant changes. Accurately predicting the Q-factor is crucial for optimizing system design and improving system stability. With the continuous advancement of machine learning (ML) techniques, their application in optimizing FSO system performance has gradually become a hot topic in the field. The authors in [6] employed deep learning for detecting signals of OOK-modulated data across different FSO channels. This approach significantly improved system performance, reduced prediction time, and lowered energy consumption. The authors in [7] estimated the refractive index structure parameter  $C_n^2$  from experimentally measured

macroscopic meteorological parameters over four quarters. They applied ML algorithms to model both the refractive index structure parameter  $C_n^2$  and the outage probability  $P_{out}$ . In reference[8], supervised machine learning techniques were used to predict the Q-factor of the FSO link, considering multiple transceivers, wavelengths, and different weather conditions as input features for the MLAs model.

This paper examines the performance of FSO systems across various link distances and weather conditions, utilizing machine learning algorithms to forecast the Q-factor. The model's prediction accuracy is assessed using MAE, RMSE, and the  $R^2$  coefficient. To predict the Q-factor of the FSO system, we trained and evaluated linear regression (LR), support vector regression (SVR), convolutional neural networks (CNN), BP neural networks, and random forest (RF) regression models. Additionally, an NRBO-BP regression prediction model is proposed, where the NRBO is used to optimize the parameters of the BP neural network. The NRBO algorithm, with its strong global search and local optimization capabilities, significantly enhances the training performance of the BP neural network[9]. Simulation results indicate that the NRBO-BP prediction model shows smaller errors compared to the original BP model, demonstrating superior prediction accuracy and good generalization ability.

## II. EXPERIMENTAL SETUP

The FSO system typically comprises three key components: the transmitter, the free-space optical channel, and the receiver, as shown in Figure 1. The transmitter system module includes a pseudo-random binary sequence (PRBS) generator, an NRZ pulse generator, a continuous wave (CW) laser, a Mach-Zehnder modulator, and optical amplifiers. The FSO channel is the medium through which the optical waves propagate between the transmitter and receiver. Due to variations in weather conditions and turbulence effects, the signal undergoes attenuation during transmission through the channel. We assume the channel is memoryless, stationary, and experiences slow fading. The received signal can be modeled using the basic channel model[10]:

$$y = RHx + N \quad (1)$$

where  $R$  is the photodetector's sensitivity.  $H$  denotes the channel state, reflecting attenuation due to atmospheric turbulence, which is caused by random fluctuations in temperature and pressure.  $N$  is the additive white Gaussian noise at the receiver.

The receiver module includes a photodetector, low-pass Bessel filter, 3R generator, and BER analyzer. The BER analyzer calculates the Bit Error Rate (BER) and Q-factor to assess the system's communication quality. The relationship between the Q-factor and BER is as follows[11]:

$$P_e = \frac{1}{2} \operatorname{erfc} \left( \frac{Q}{\sqrt{2}} \right) \quad (2)$$

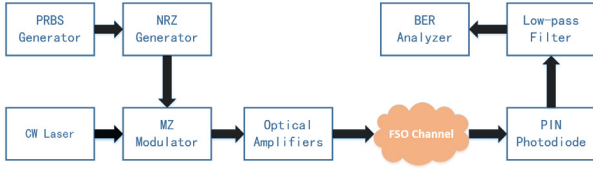


Figure 1. FSO system block diagram

The FSO system was simulated using OptiSystem, with the system design parameters provided in Table I. Performance analysis of the FSO system was conducted under various link distances and weather conditions. Figures 2 and 3 show the variations in the Q-factor and received signal power with respect to link distance (Spanning from 0.1 to 5 kilometer) under different weather conditions. The results indicate that as the link distance increases, both the received signal power and the Q-factor decrease, with the FSO system performing worst in foggy conditions.

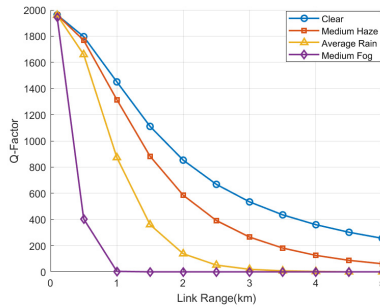


Figure 2. Q-factor against link range at 1550nm wavelength for different weather conditions

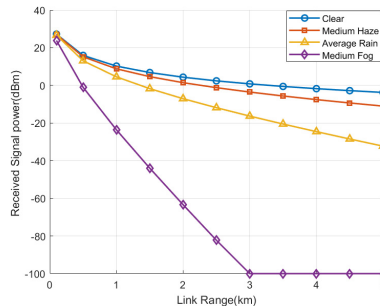


Figure 3. Received signal power against link range at 1550 nm wavelength for different weather conditions

TABLE I. FSO SYSTEM PARAMETERS

| Parameter         | Value         | Unit |
|-------------------|---------------|------|
| Transmission rate | 10            | Gbps |
| Optical power     | 5-25          | mW   |
| Wavelength        | 850,1310,1550 | nm   |
| Link range        | 100-5000      | m    |
| Beam divergence   | 2-5           | mrاد |

### III. ML MODEL EVALUATIONS

#### A. Dataset Generation

This section applies machine learning regression algorithms to estimate the Q-factor of the proposed FSO system. The dataset is generated using OptiSystem software via parameter scanning, producing approximately 3824 data points for the input parameters. These input parameters include transmission power, wavelength, link distance, beam divergence, and atmospheric attenuation under different weather conditions, with the Q-factor as the output. The dataset is split into two parts: 70% is used for training, and the remaining 30% for testing.

#### B. NRBO-BP Model

The Newton-Raphson-Based Optimizer (NRBO) is a novel intelligent optimization algorithm with strong global search and local optimization capabilities. Inspired by the Newton-Raphson method, NRBO explores the entire search process using two key rules: the Newton-Raphson Search Rule (NRSR) and the Trap Avoidance Operator (TAO). NRSR enhances the exploration capability of NRBO and accelerates convergence, allowing it to reach improved positions in the search space. TAO helps NRBO avoid falling into local optima. This study applies the NRBO algorithm to optimize the BP network parameters, identifying the optimal weights and biases. The root mean square error (RMSE) calculated for each network training set serves as an approximate measure of fitness, which is then used to evaluate the overall fitness. The essential steps involved in the NRBO algorithm are outlined below:

- (1) Determine the number of particles  $N_p$ , the maximum number of iterations  $MaxIt$ , the decision factor  $DT$ , the search space range  $[LB, UB]$
- (2) Initialize the optimal solution, determine the best and worst positions based on fitness values, and set the best particle with the optimal fitness and position.
- (3) Main loop: Iterate until  $MaxIt$ , gradually decreasing the step size. The formula for  $\delta$  is:

$$\delta = \left( 1 - \frac{2 * Iter}{MaxIt} \right)^5 \quad (3)$$

- (4) Traverse each particle and perform the following steps:
  - a) Select two random particle and calculate their difference  $\rho$  to adjust the search direction of the current particle.

- b) Calculate the difference between the particle's current position and the optimal position. Update the particle's position based on  $\rho$  and NRSR.
- c) Based on the random factor, apply the TAO to introduce perturbation and avoid local optima. Ensure the particle's position stays within the boundaries of  $[LB, UB]$ .
- (5) Evaluate the updated particle fitness; if better, update the position and fitness.
- (6) The algorithm stops at *MaxIt* and outputs the optimal solution.

The BP neural network is primarily composed of three layers: the input layer, the hidden layer, and the output layer. These layers are interconnected, with the input layer receiving the initial data, the hidden layer performing intermediate computations, and the output layer producing the final predictions or results based on the processed information. Traditional BP algorithms use gradient-based local search, often getting stuck in local optima. This paper replaces gradient descent with NRBO, combining global search and efficient local search through Newton-Raphson, TAO, and random perturbation, enabling fast convergence and avoiding local optima. The NRBO-BP algorithm flow is shown in the Figure 4:

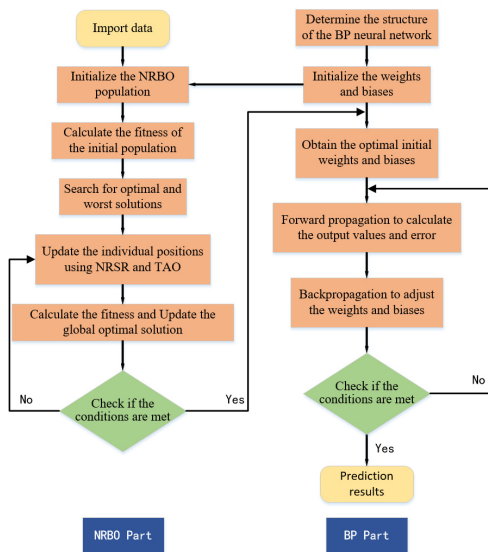


Figure 4. Flowchart of the NRBO-BP algorithm

#### IV. RESULT VERIFICATION AND ANALYSIS

The experimental environment in this study is configured with an AMD Ryzen 7-6800H processor and an Nvidia RTX-3060 GPU. The software platform utilized is MATLAB 2023b, with Visio used for plotting. The BP neural network comprises an input layer with 5 neurons, corresponding to the dimensions of the input features. A single hidden layer is employed, containing 10 neurons, where the activation function is set to

ReLU to improve the nonlinear fitting capability of the network. The output layer consists of one neuron to perform regression prediction of the Q factor, using a linear activation function. The network is trained using the mean squared error (MSE) as the loss function, with the Adam optimizer applied and the learning rate configured to 0.001.

This paper compares the NRBO-BP model with LR, SVR, CNN, BP neural networks, and RF models. The parameter settings for the NRBO-BP model are shown in Table II. To accurately reflect the actual bias of the proposed model's prediction error, three error metrics are introduced:  $R^2$  coefficient, MAE and RMSE. Specifically, the reference formulas are as follows[12]:

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (4)$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (5)$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (6)$$

Where  $n$  is the quantity of sample,  $y_i$  is the actual value of the variable,  $\hat{y}_i$  is the forecasted value of the variable, and  $\bar{y}$  is the mean of the actual values.

TABLE II. PARAMETER SETTINGS OF NRBO-BP ALGORITHM

| Parameter Type                        | Values    |
|---------------------------------------|-----------|
| Number of neurons in the input layer  | 5         |
| Number of neurons in the hidden layer | 10        |
| Learning rate                         | 0.001     |
| Target error                          | $10^{-6}$ |
| Number of BP training iterations      | 1000      |
| Number of samples                     | 3824      |
| Population size                       | 10        |
| Decision Factor for TAO               | 0.6       |

TABLE III. PERFORMANCE COMPARISON OF DIFFERENT MODELS

| ML Model       | $R^2$        | MAE           | RMSE          |
|----------------|--------------|---------------|---------------|
| LR             | 0.387        | 155.559       | 250.164       |
| SVR            | 0.706        | 151.140       | 173.336       |
| CNN            | 0.968        | 23.997        | 56.706        |
| BP             | 0.971        | 26.403        | 54.833        |
| RF             | 0.960        | 22.651        | 64.157        |
| <b>NRBO-BP</b> | <b>0.992</b> | <b>14.151</b> | <b>29.390</b> |

To accurately predict the performance of FSO links, different ML models were studied and compared with the proposed model. As shown in Table III, the NRBO-BP model performs the best in terms of  $R^2$ , MAE, and RMSE, indicating that this model has higher prediction accuracy and performance advantages. It can be seen that LR and SVR performed the worst, with  $R^2$  values of 0.387 and 0.706, respectively. Therefore, these two models are unsuitable for estimating the output Q-factor. The proposed NRBO-BP model achieved an  $R^2$  of 0.992, very close to 1, indicating excellent data fitting and strong predictive capability. Additionally, the NRBO-BP model also provides robust results in terms of MAE and RMSE. In comparison to the traditional CNN model, BP model, and RF model, the NRBO-BP model shows a reduction in both MAE and RMSE. Specifically, the MAE decreases by 41.01%, 46.38%, and 37.54%, while the RMSE is reduced by 48.17%, 46.38%, and 54.19%. This demonstrates that the proposed model outperforms BP in terms of prediction accuracy and optimization ability, offering more precise predictions and better generalization.

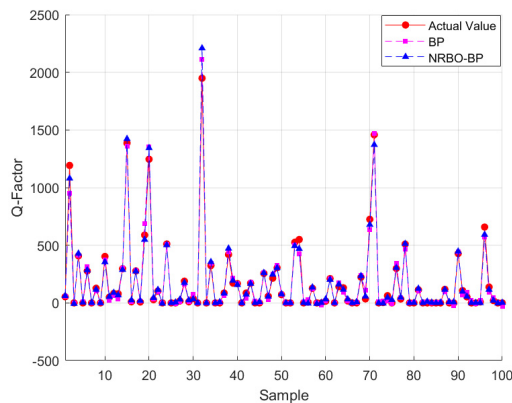


Figure 5. Comparison of prediction results of different models

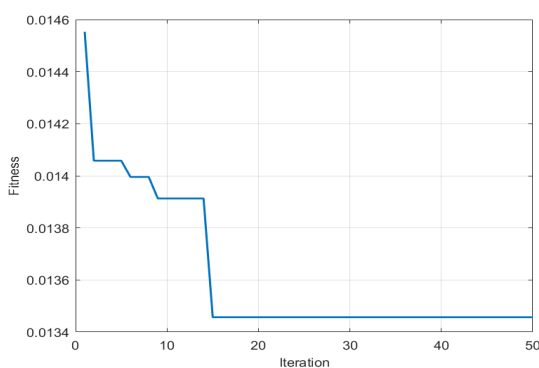


Figure 6. Fitness curve in Q-factor prediction

Figure 5 shows the comparison between the predicted Q-factor values from the BP and NRBO-BP models and the actual values for the first 100 samples in the test set. The results show that the proposed NRBO-BP model can more accurately predict FSO link performance and better fit the data curves. Figure 6

presents the fitness curve for Q-factor prediction. It can be observed that the algorithm converges quickly in the early stages. After approximately 20 iterations, the curve stabilizes, indicating that the algorithm has reached a steady solution.

The experimental results indicate that the NRBO-BP algorithm outperforms the BP neural network in prediction. However, some peak points exhibit poor fitting, which may be due to the BP simulation process and its network structure.

## V. CONCLUSIONS

In this work, performance analysis of the FSO system was conducted under different link ranges and weather conditions. Machine learning models, including LR, SVR, CNN, BP neural networks, and RF, were used to estimate the Q-factor of the FSO system. The NRBO algorithm was also applied to optimize the parameters of the BP neural network, leveraging NRBO's strengths in global optimization search. Experimental results demonstrate that the NRBO-BP model outperforms others in terms of  $R^2$ , MAE, and RMSE, showcasing strong predictive and generalization capabilities in FSO link quality estimation. Furthermore, the algorithm provides valuable insights for predictions in other domains.

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